

EBAU: EKAINA 2018

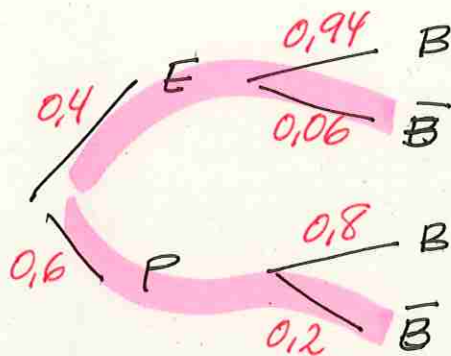
A3

$E \equiv$ aurreseki emon kreditua.

$P \equiv$ partikulari emon kreditua

$B \equiv$ Berresturatu dimo

$\bar{B} \equiv$ Ez berresturatu.

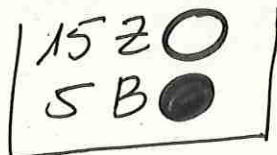


$$\begin{aligned} a) P(\bar{B}) &= P(E \cap \bar{B}) + P(P \cap \bar{B}) = \\ &= P(E) \cdot P(\bar{B}/E) + P(P) \cdot P(\bar{B}/P) \\ &= 0.4 \cdot 0.06 + 0.6 \cdot 0.2 = \\ &= \boxed{0.144} \quad \text{Prob. osoa} \end{aligned}$$

$$b) P(E/\bar{B}) = \frac{P(E \cap \bar{B})}{P(\bar{B})} \quad \text{Bayes ("a posteriori")}$$

$$P(E/\bar{B}) = \frac{P(E) \cdot P(\bar{B}/E)}{P(\bar{B})} = \frac{0.4 \cdot 0.06}{0.144} = \boxed{0.166}$$

B3



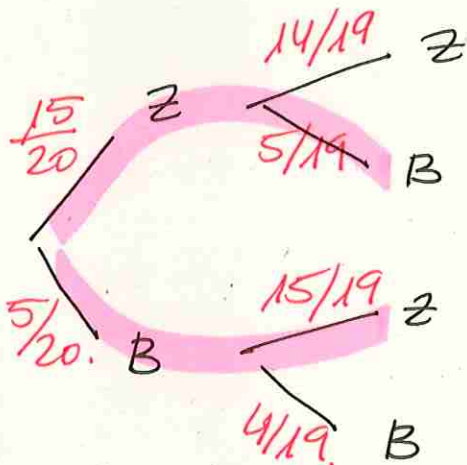
$Z \equiv$ zuria (zatea)

$B \equiv$ beltza (zatea)

$$a) P(Z) = 15/20 = 3/4 = 0.75 \quad \text{LAPLACE}$$

$$b) \text{ Bi atzaro eto biak zatezaren prob. (neurpeko prob.)}$$

$$P(Z \cap Z) = P(Z_1) \cdot P(Z_2/Z_1) = \frac{15}{20} \cdot \frac{14}{19} = \frac{21}{38} = 0.552$$



c) Leheneuzko beltza izan bada bigarren beltza ere zatezaren prob.

$$P(B_2/B_1) = 4/19 = 0.21$$

d) Kolore desberdinetakoa.

$$\begin{aligned} P(\text{desberdina}) &= P(B \cap Z) + P(Z \cap B) = \\ &= P(Z) \cdot P(B/Z) + P(B) \cdot P(Z/B) = \\ &= \frac{15}{20} \cdot \frac{5}{19} + \frac{5}{20} \cdot \frac{4}{19} = 0.394 \end{aligned}$$